

Integration of Power Generation and Oil Refining

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DM2

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Introduction

- Most basic need today → Energy
- Increasing demands and price of oil comes an increased environmental consciousness
- Justifies exploration of costly yet cleaner methods of power production

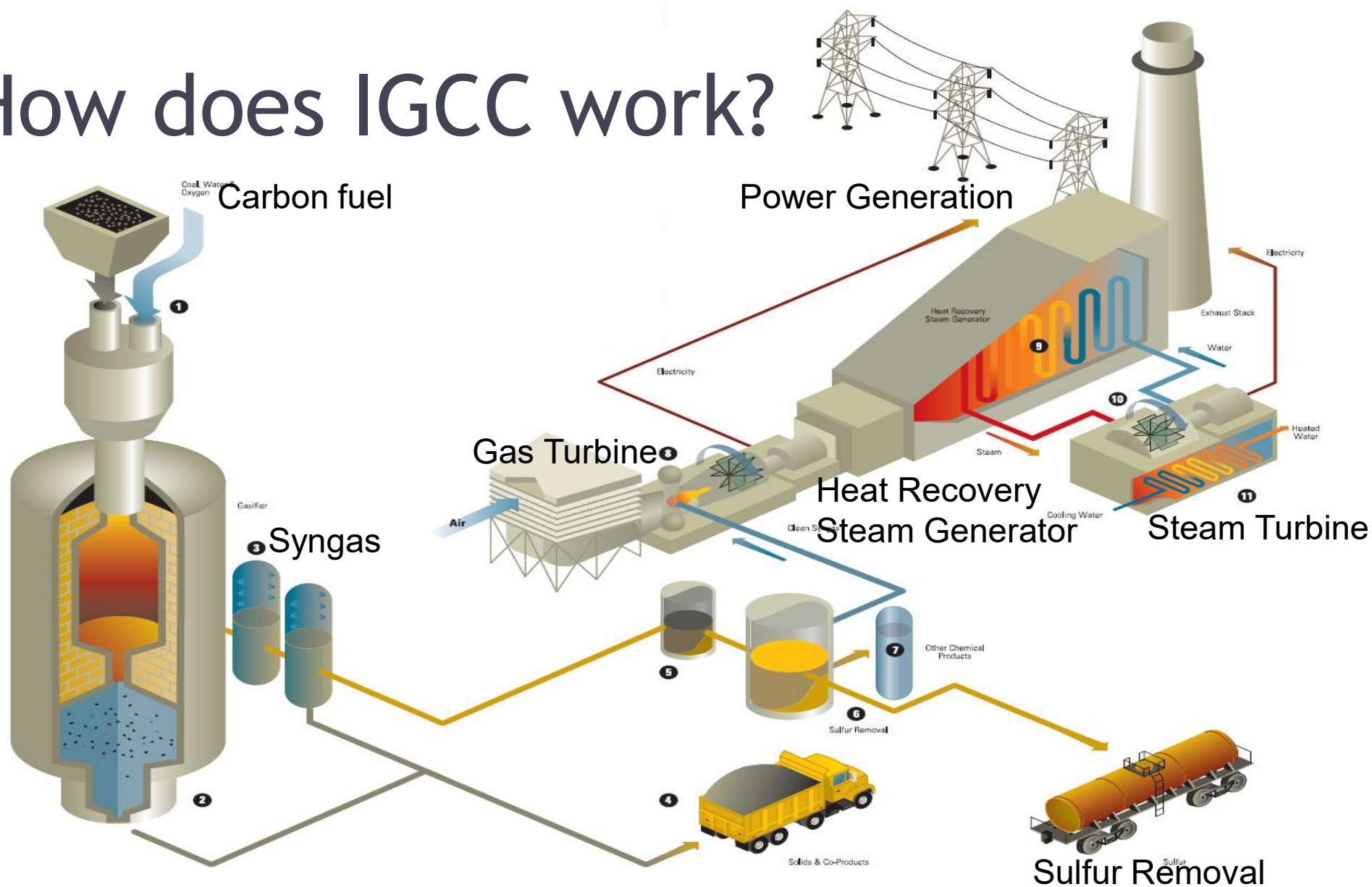


Introduction

- One solution is to integrate a power plant into an existing oil refinery
- Integrated Gasification Combined Cycle (IGCC)
- Important for world energy market
- Low cost feed stocks can produce low-cost electricity while meeting environmental regulations



How does IGCC work?



Objectives

- To evaluate the feasibility of redesigning an oil refinery, AUR, into an IGCC plant
- To compare an alternative method to produce power through the gasification of short resid (instead of delayed coke)
- To eliminate CO₂ emissions through implementation of a sequestration system

Overall AUR Plant

ISBL

Conversion Section

- Coker
- Vacuum Unit

Syngas

- Gasification of Delayed Coke

Steam Cracking

OSBL

Power Generation

- Coal-fired fluidized bed boilers
- Steam Turbines
- 540 MW

Other Offsites

- Tankage/Storage
- Cooling Water
- Site Development

Overall DM2 Plant

ISBL

**Conversion
Section**

- Vacuum Unit

Syngas

- Gasification of Short Resid

**Steam
Cracking**

OSBL

Power Generation

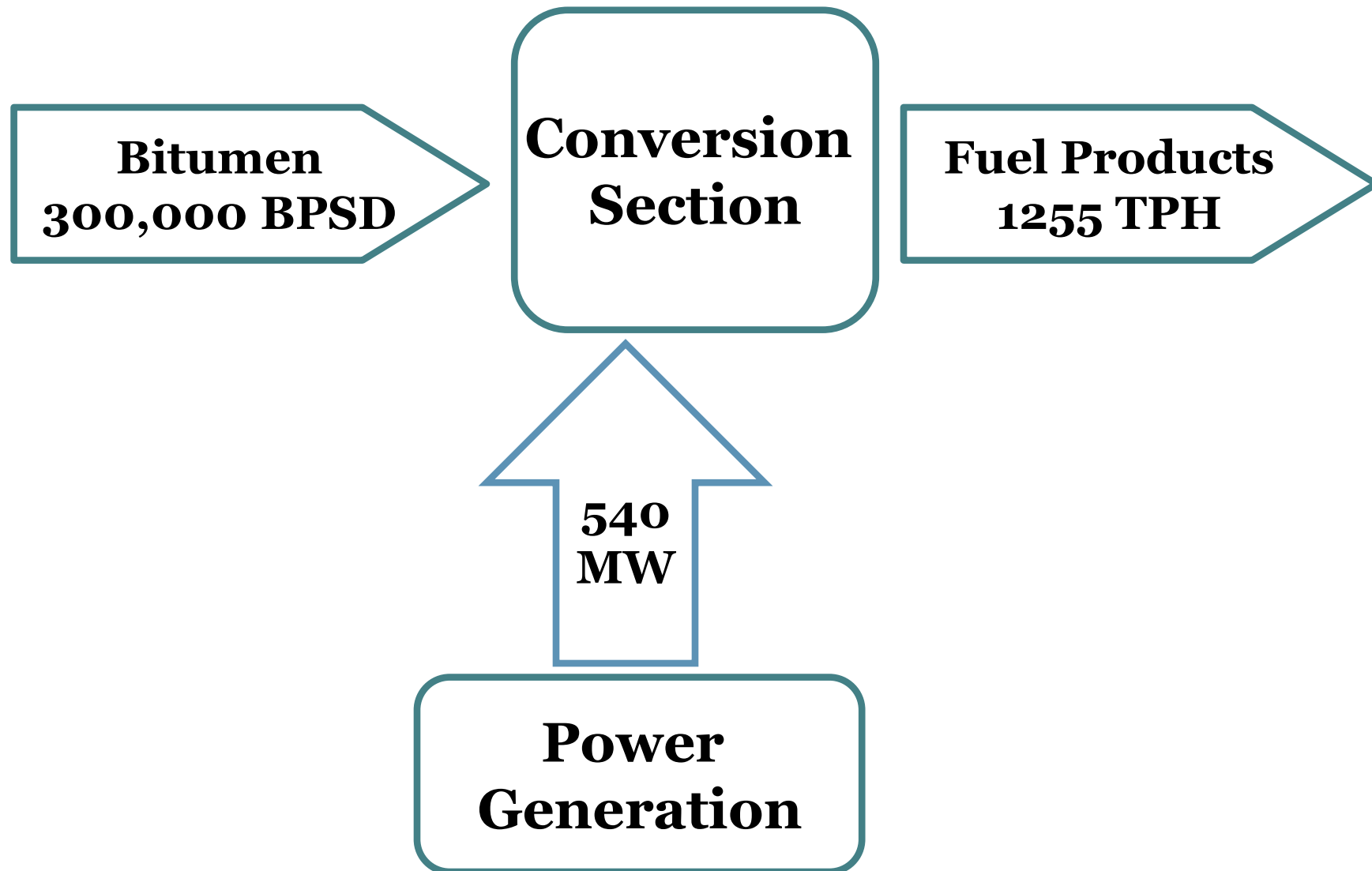
- Steam Turbines
- Gas Turbines
- 5000 MW

Other Offsites

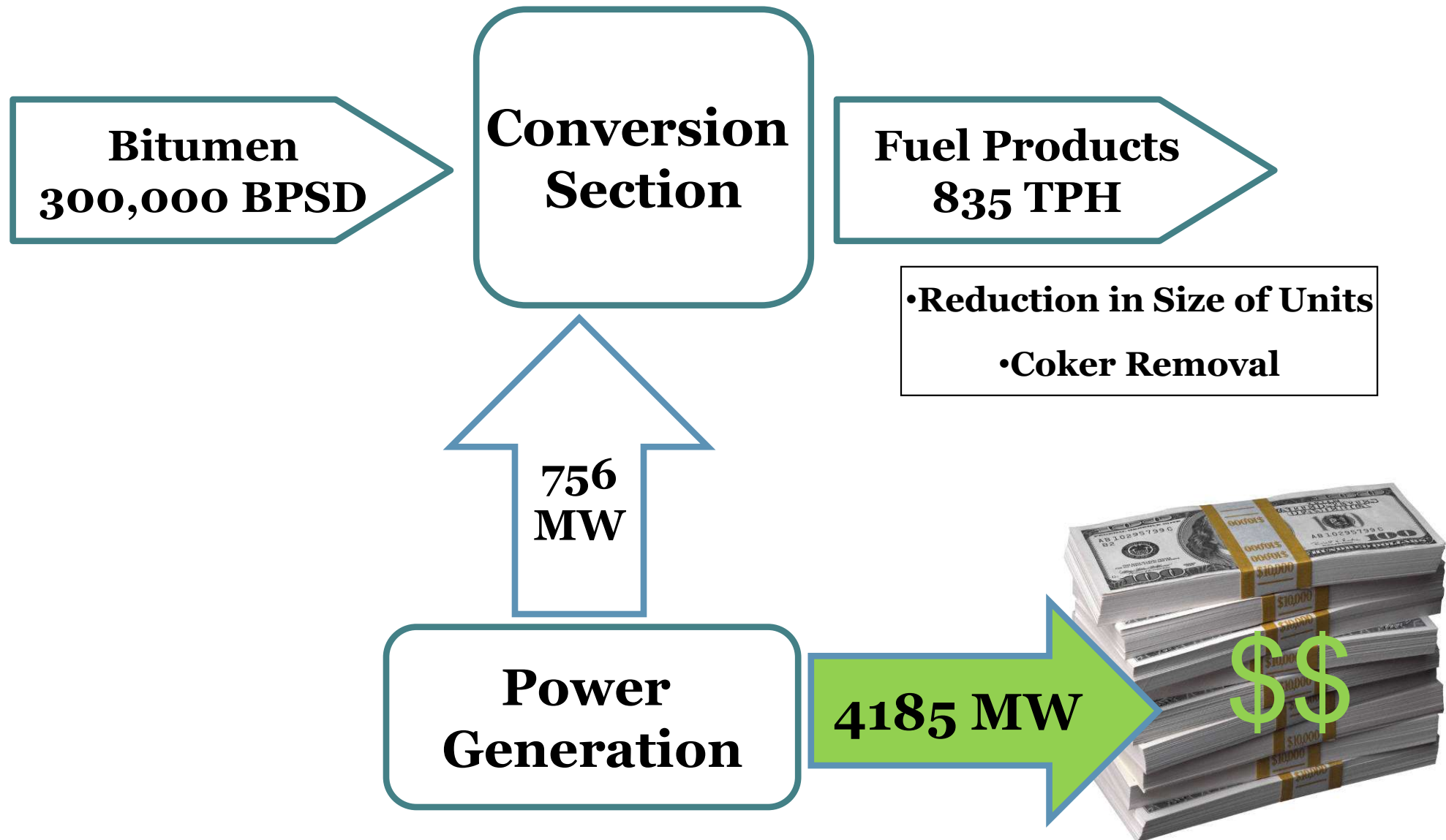
- Tankage/Storage
- Cooling Water
- Site Development

Reduction in Size of Units

Overall AUR Plant



Overall DM2 Plant

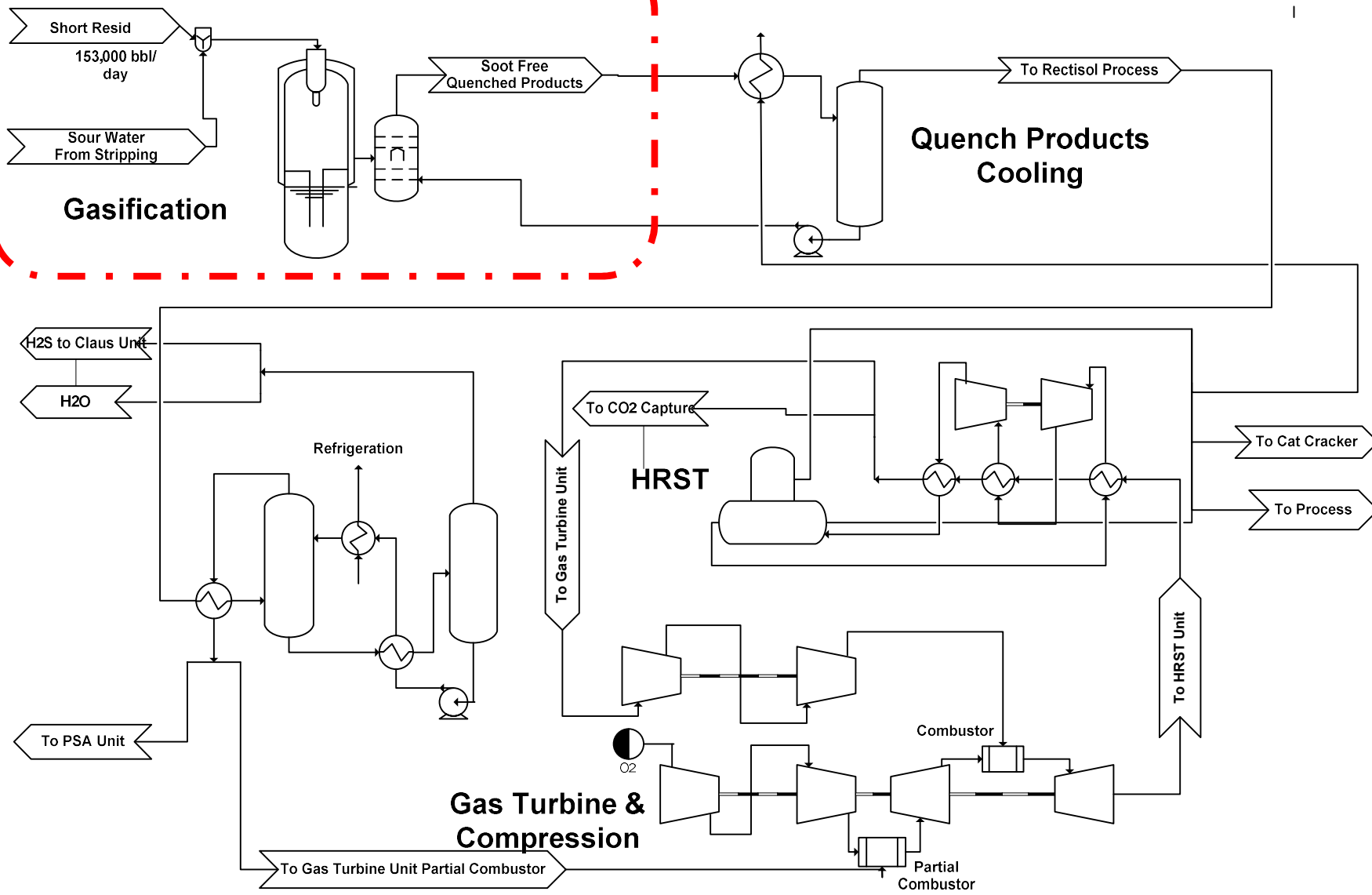


Gasification

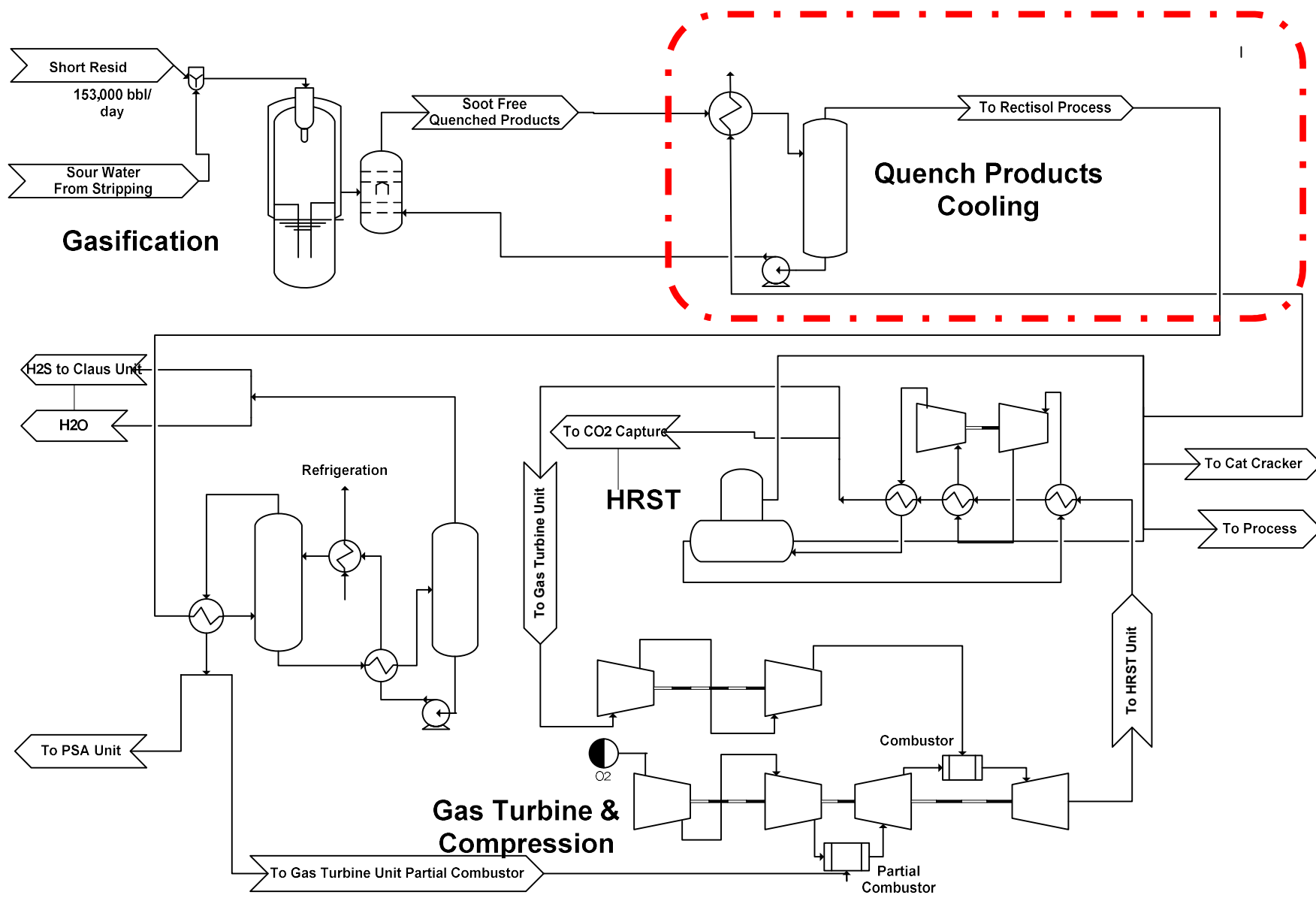
Short Resid
153,000 bbl/day

Sour Water From Stripping

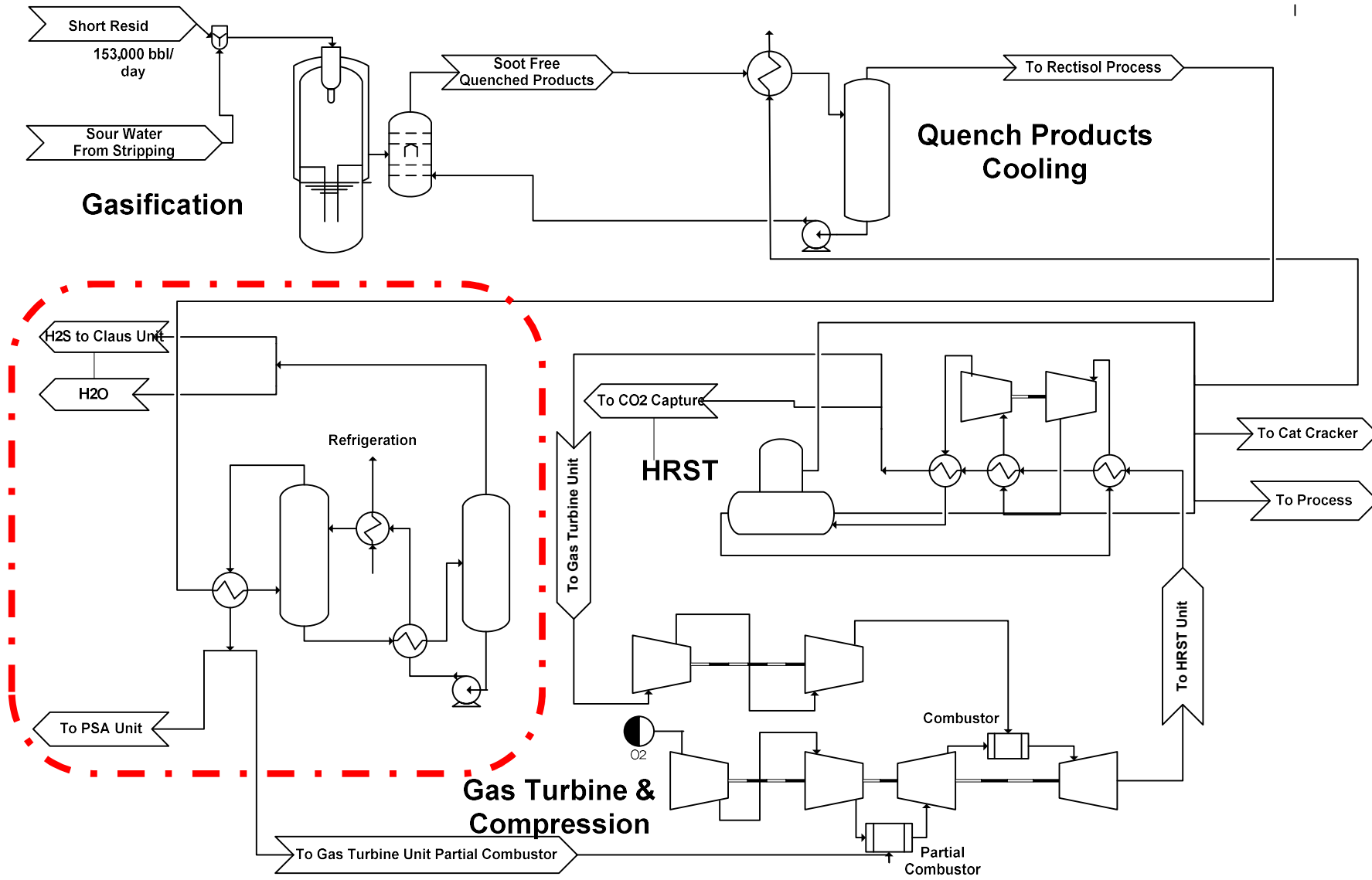
Soot Free Quenched Products



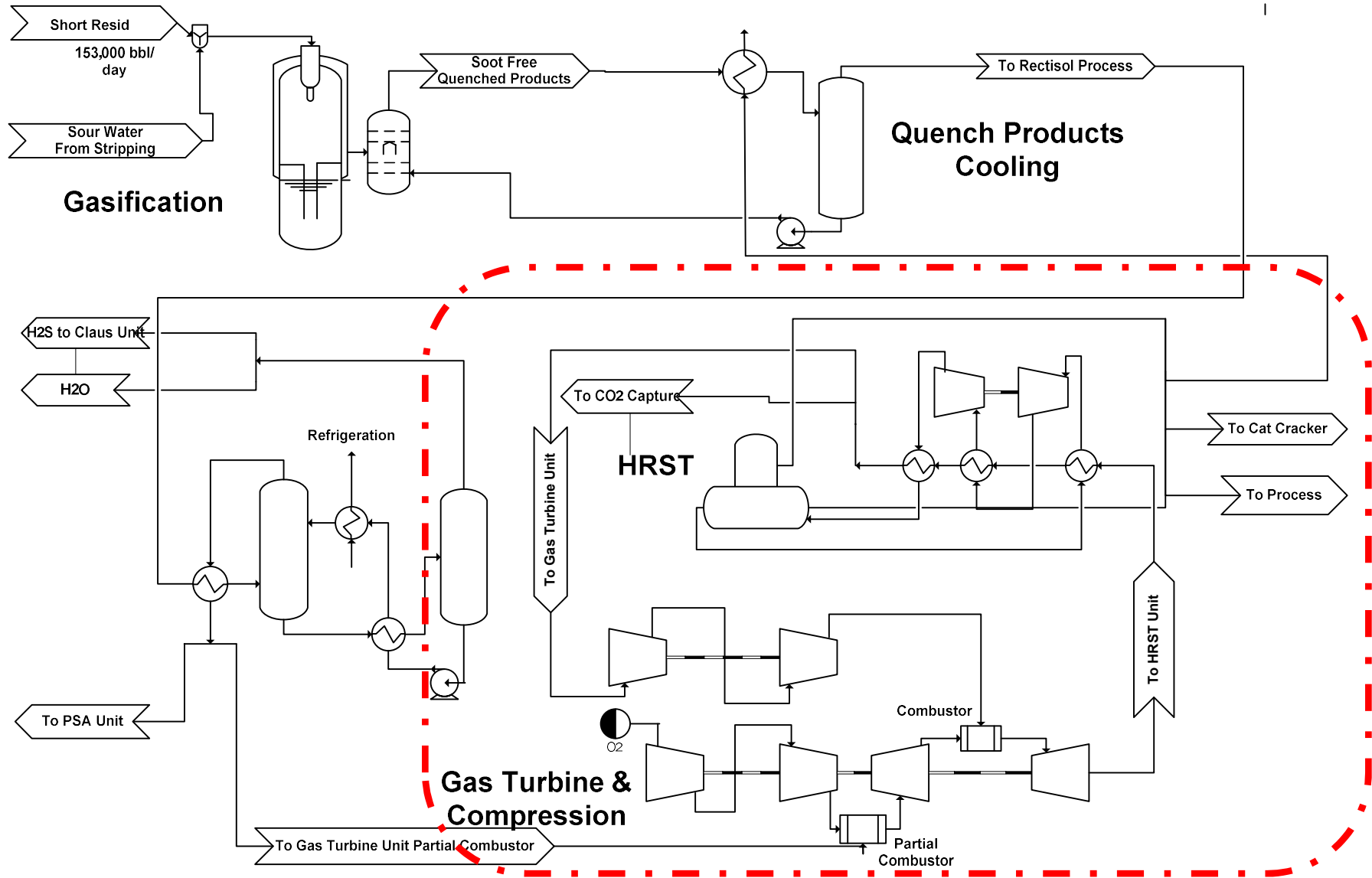
Process Flow Diagram



Process Flow Diagram



Process Flow Diagram



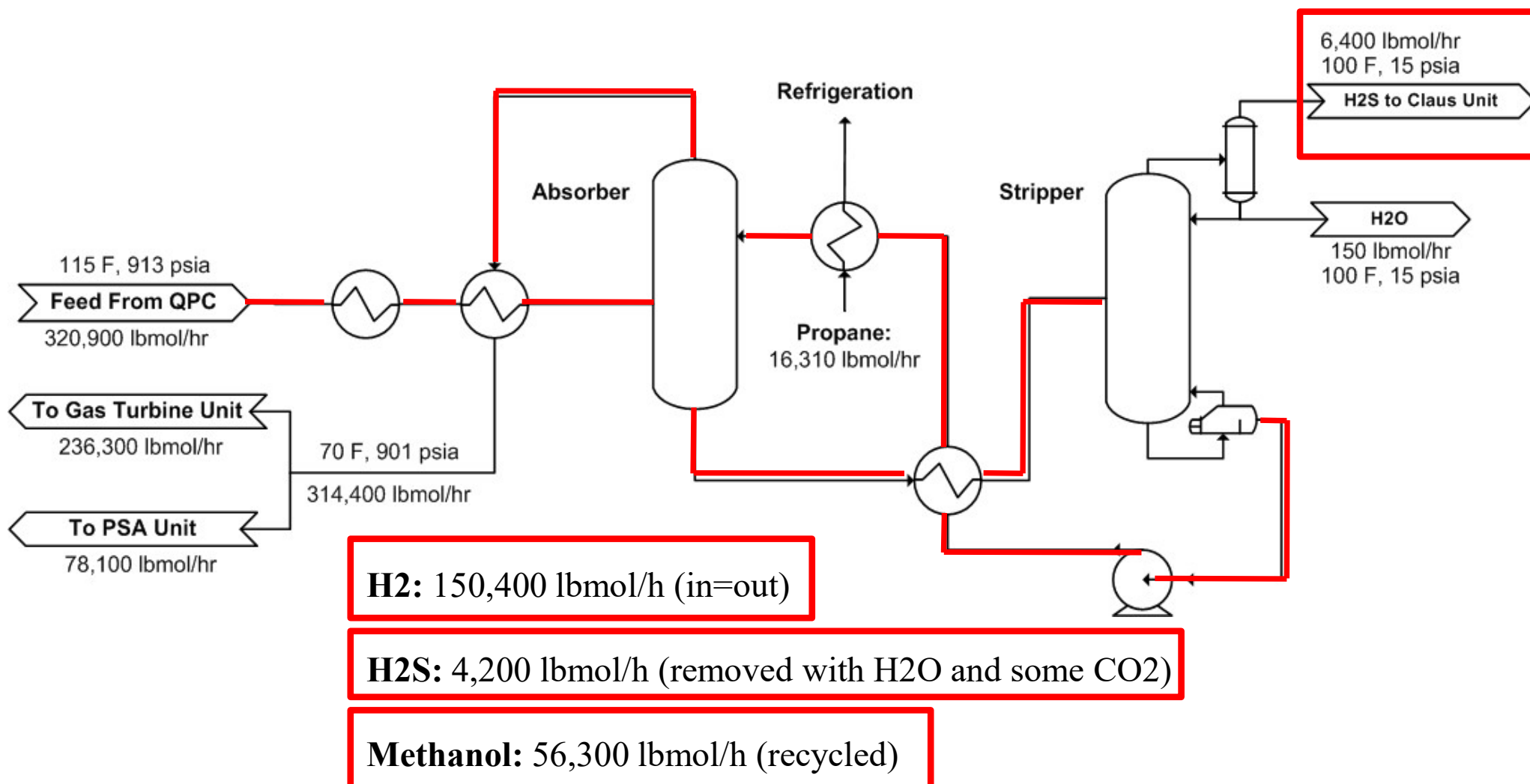


Refining of Syngas

Rectisol Process

- Rectisol Process removes H_2S from gas sent to Gas Turbine and Compression unit.
- Achieved using Methanol loop
- Methanol has a high selectivity towards H_2S and a deselectivity towards H_2 . – dependent on temperature
- To provide adequate cooling (desired temperature) for Rectisol process, a propane refrigeration loop was utilized.

Rectisol and Refrigeration - PFD





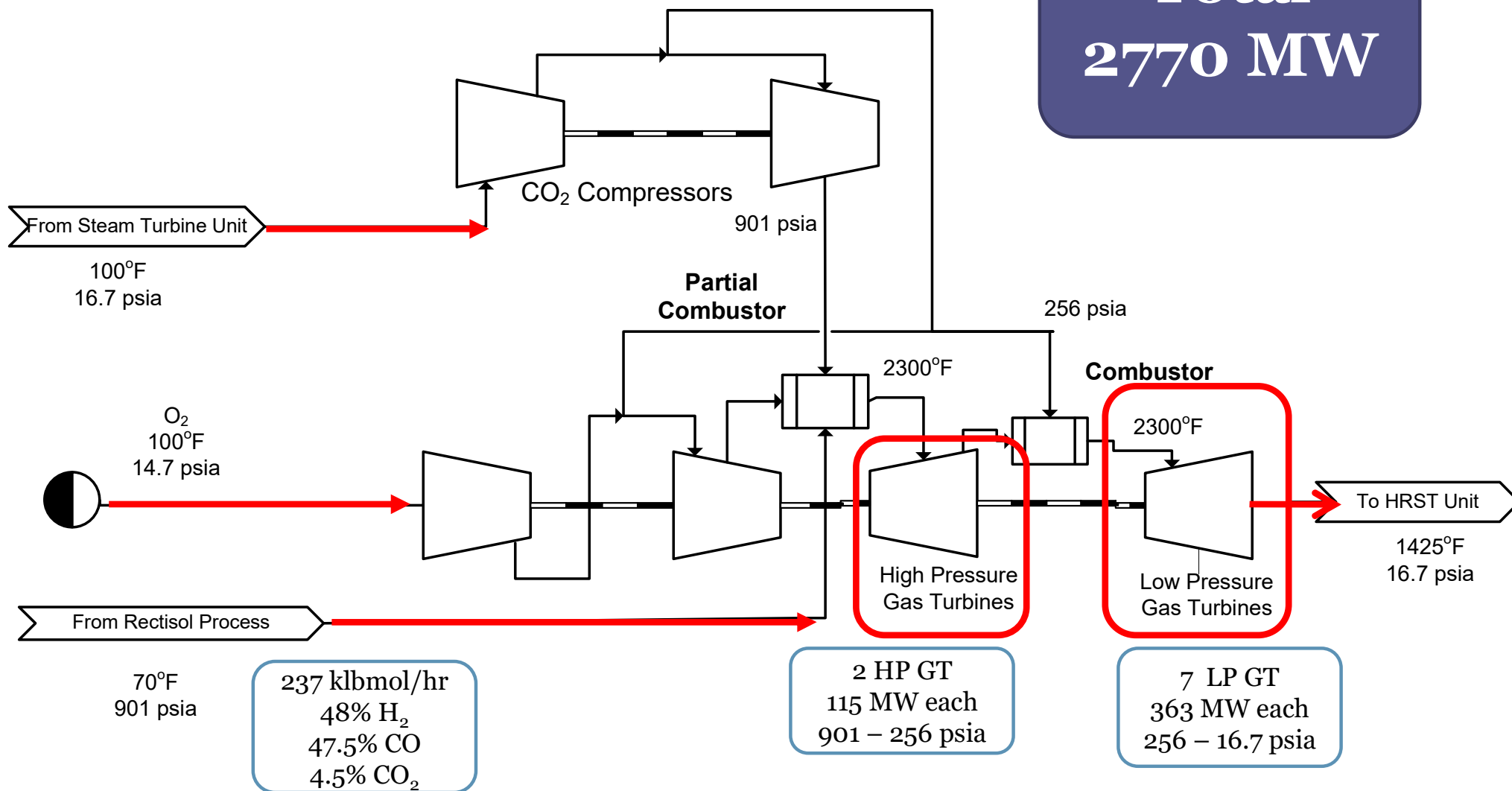
Power Generation

Power Generation

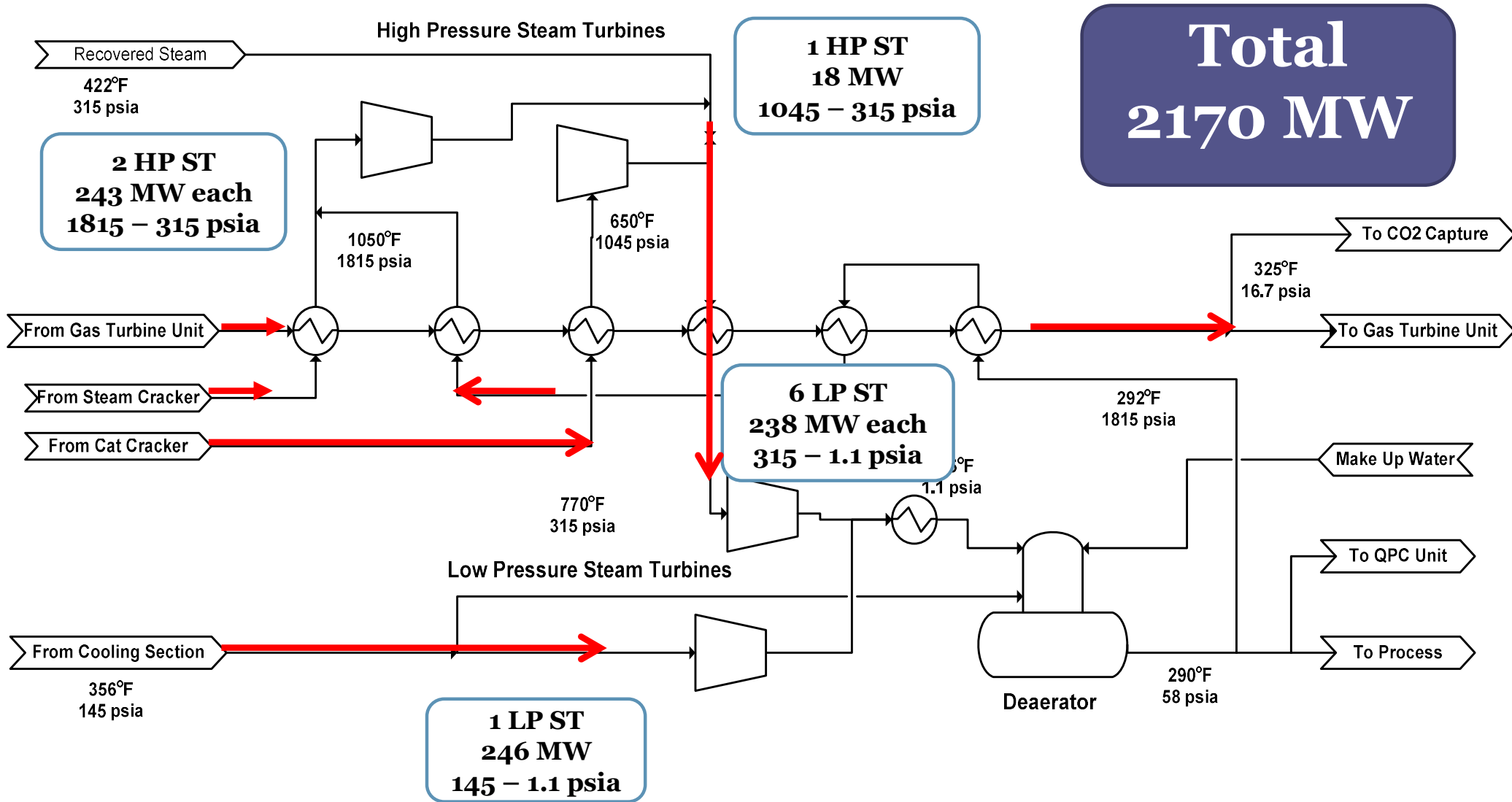
- Net power generation of DM2 is nearly 8 times larger than that of AUR
- Redesigned Power Plant:
 - Gas Turbine and Recycle CO₂ Compression
 - Heat Recovery and Steam Turbine

Gas Turbine & Compression

**Total
2770 MW**

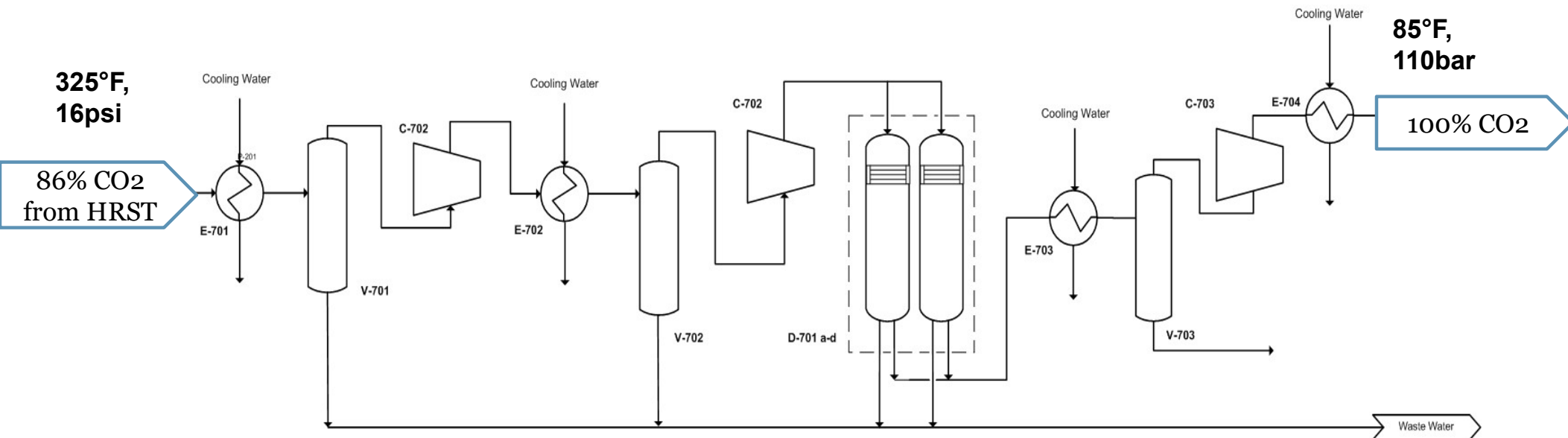


Heat Recovery and Steam Turbines



CO₂ Sequestration

275 MW



Pipeline Requirements:
Liquid, 110 bar, 85°F, 10 ppm H₂O

Power Generation Costs

Equipment	Installed Cost (Million\$)
Turbine and Power Units	\$1,651
Gasification and Cooling	\$2,377
Rectisol Process	\$62
Compressors for Sequestration	\$301
Total of Power Generation Units	\$4,391
Plus 40% Contingency	\$6,147

Gas
Turbines
\$785

Steam
Turbines
\$645

Total installed cost of power plant: \$6.1 billion
\$5.2 billion more than AUR cost of \$0.83 billion

Overall Energy Requirements

Total Energy Produced/Required(MW)		
Unit	AUR	DM2
ASU/O ₂ Comp.	-190	-156
Gasification and Rectisol	-76	-47
Delayed Coker	-26	0
Other	-192	-192
CO ₂ Compression	0	-275
Additional Pumps	0	-31
Generator and Distribution Losses	-20	-20
Total Energy Required	-540	-756
Gas Turbines	0	2770
Steam Turbines	540	2171
Total Energy Produced	540	4941
Net Energy Produced	0	4185

- Net Power Generation by DM2: 4185 MW

- Electricity sale price: \$0.07/kWh

- \$2.4 billion in electricity credit



Economic Analysis

Power Generation Costs

Equipment	Installed Cost (Million\$)	
Gas Turbines	\$785	{ For the Gas and Steam Turbine Units }
Steam Turbines	\$645	
Pumps	\$2	
Heat Exchangers	\$4	
Vessels	\$3	
Turbine and Power Units		\$1,651
Gasification and Cooling		\$2,377
Rectisol Process		\$62
Compressors for Sequestration		\$301
Total of Power Plant		\$4,391
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Gas Turbines \$785	Steam Turbines \$645
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Economics - Comparison of Costs

Comparison of Power Generating Facility of Proposed Process and <i>AUR</i>		
	Proposed Process	<i>AUR</i>
Capital Costs with 40% Contingency (Million\$US)	\$6,147	\$826
Total Power Output (MW)	4,185	540
Capital Cost / kW (\$/kW)	\$1,470	\$1,530
Total CO2 Emitted (Metric Tons/day)	0	11,070

Economics - Comparison of Costs

Project Economics		
Parameter	DM2	AUR
Capital Cost per kW for the Power Plant (\$USD/KW)	\$1,470	\$1,530
OPEX per KWh for the Power Plant (\$USD/KWh)	\$0.12	\$0.04
Internal Rate of Return (\$50 Carbon Tax)	12.8%	19.9%
Break-Even Carbon Tax (\$USD/Metric Ton/Day)	\$115	
Break-Even Internal Rate of Return	12.4%	

Economics - Comparison of Costs

Comparing Annual Cash Flows		
	Proposed Process	AUR
Labor, Maintenance, Insurance & Other Royalties	\$1,044	\$614
Fuel Cost	\$2,384	\$13
Utilities Cost (Steam, Water & Oxygen)	\$562	(Included in Raw Mat.)
Raw Materials	\$218	\$2,602
Total Plant Operating Costs (Million\$US)	\$4,208	\$3,229
Product Yields (Kerosene, Gasoline, Hydrogen, etc.)	\$4,197	\$6,344
Electricity Sold to Grid	\$2,415	\$0
CO2 (\$50 Carbon Tax)	-\$56	-\$699
H2 Sold to PSA Unit	\$186	\$0
Total Plant Revenues (Million\$US)	\$6,741	\$5,645
TOTAL ANNUAL CASH FLOW (Million\$US)	\$2,533	\$2,416

Conclusion

- Based on IRR values, the proposed process is not economically feasible compared to AUR
- Major contributors of annual profit include cost of fuel, electricity generated and carbon tax, if implemented
- A lower cost of fuel could enhance the profitability of this design

Thank You

Special Thanks to:

Dr. David McKeagan

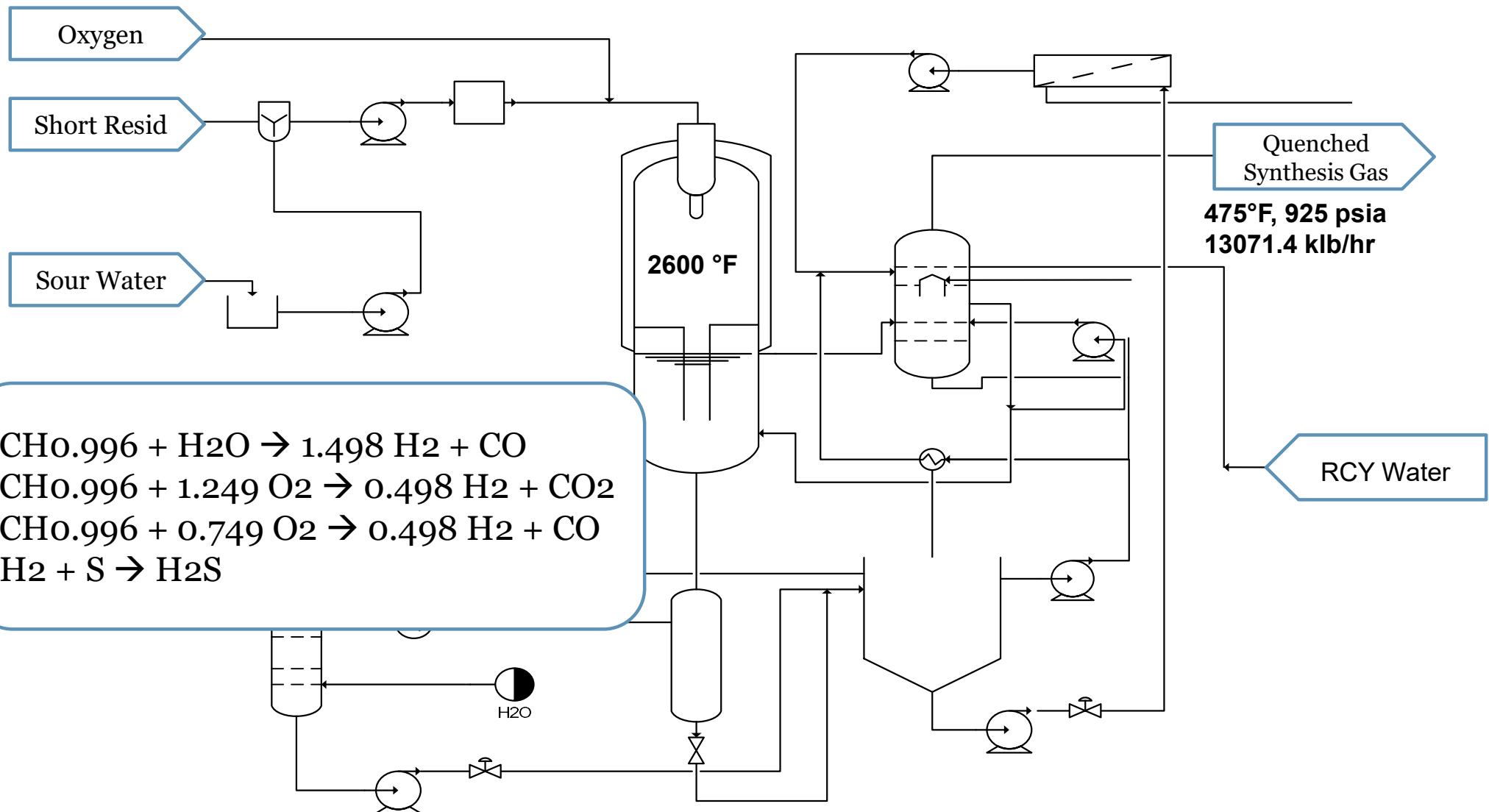
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- Walas, Stanley M., *Chemical Process Equipment: Selection and Design*, USA ButterWorth-Heinemann, 1990
- “IGCC Cycle,” www.aep.com/aboutigcc (Apr. 2008)

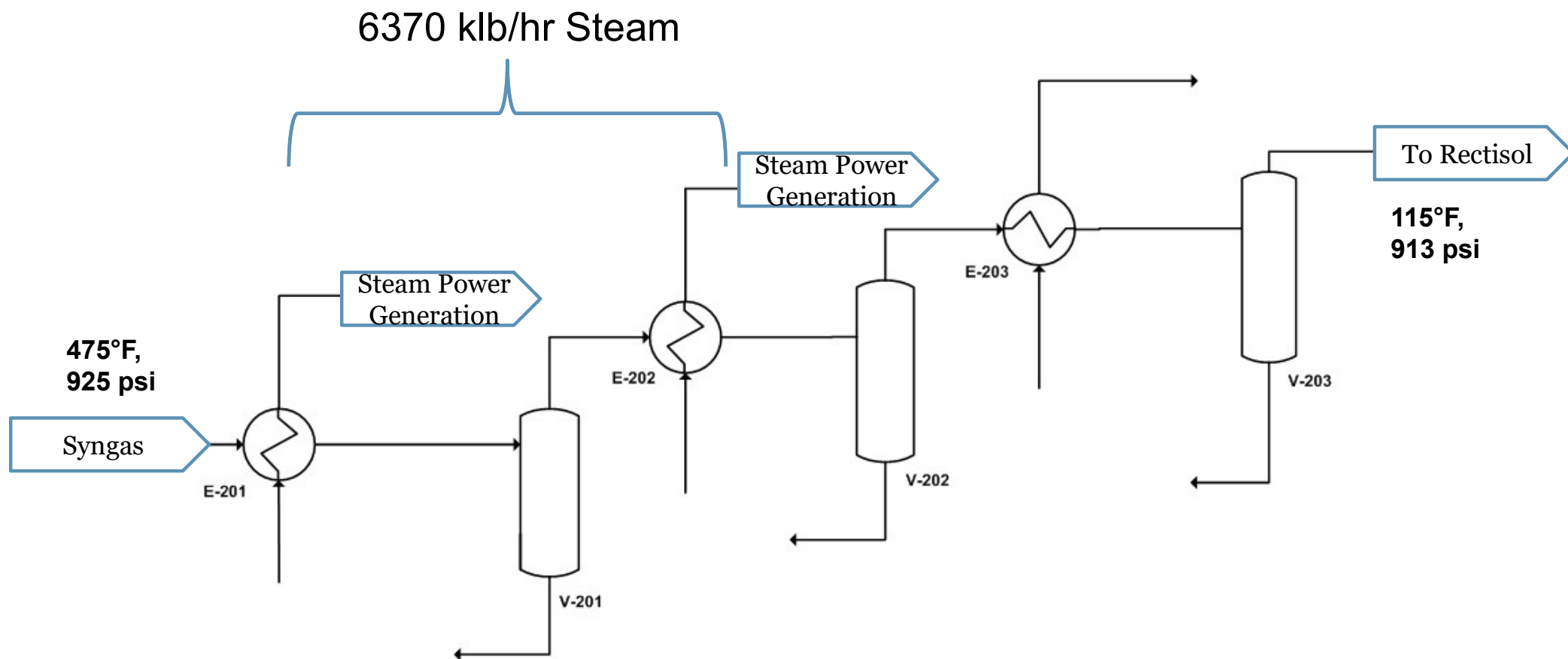


Extra Slides

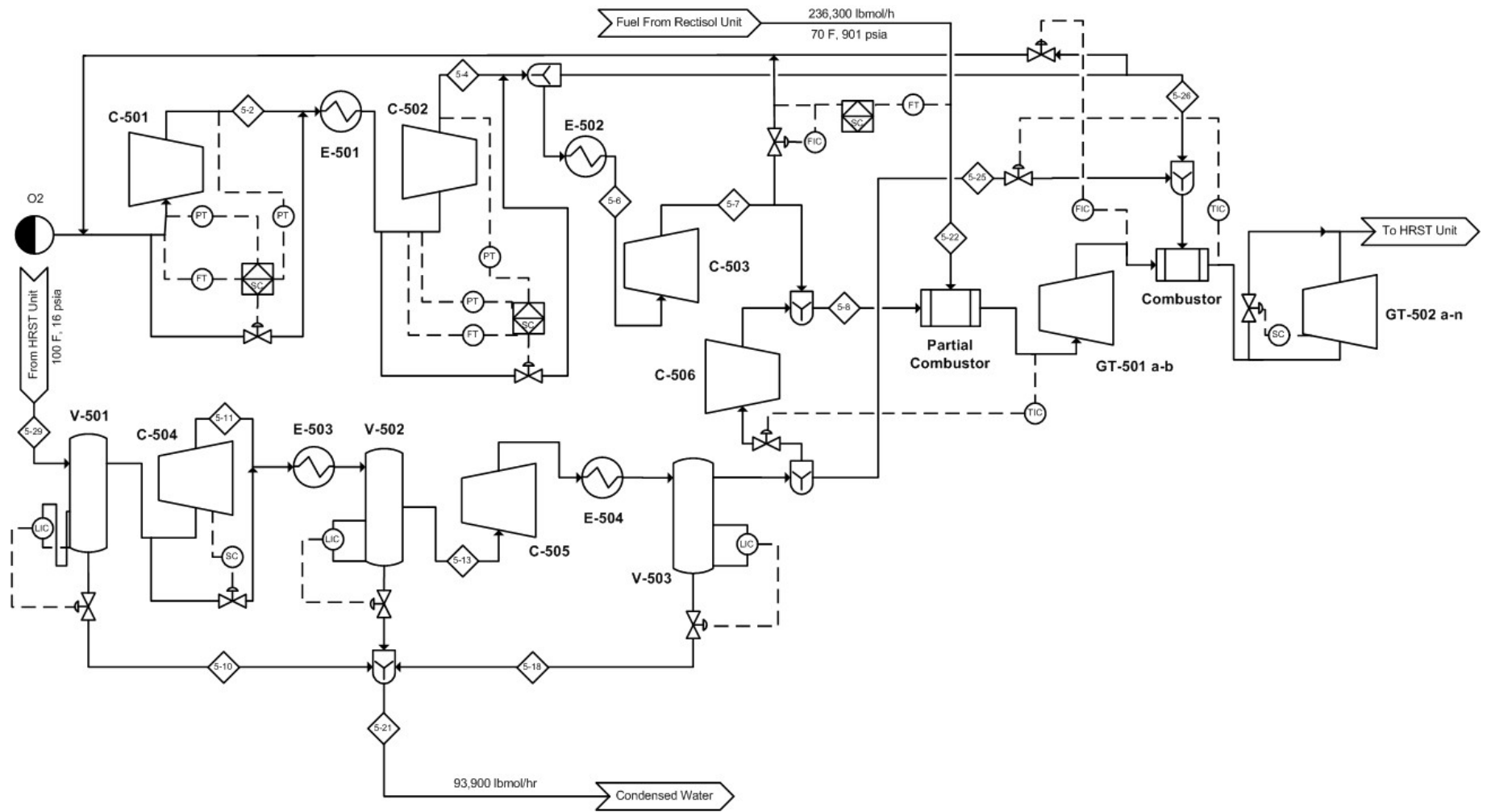
Gasification



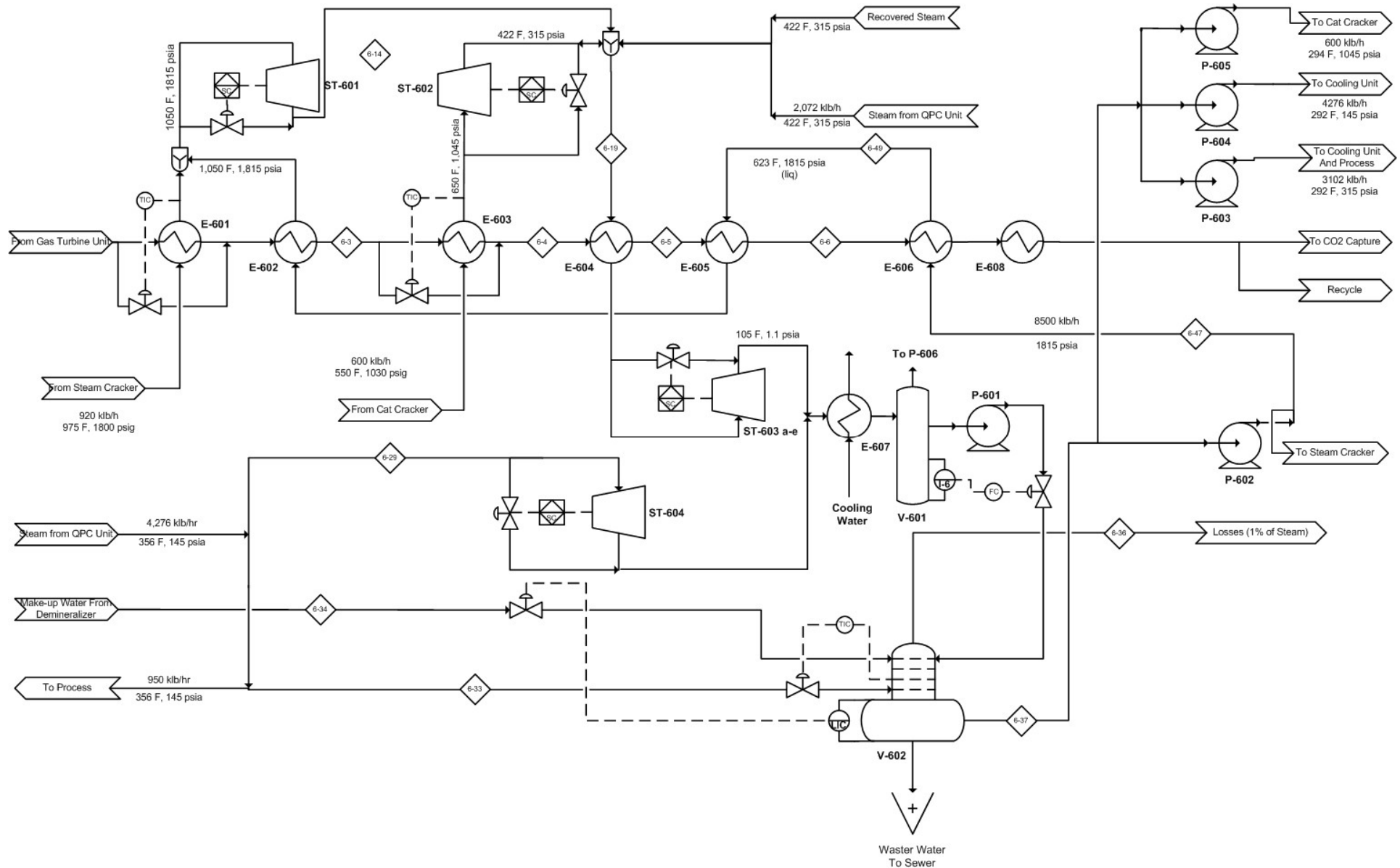
Quenched Products Cooling Section



Gas Turbine and Compression: PFD



Heat Recovery and Steam Turbines: PFD



Turbine Summary

Turbine Description	Number of Turbines	MW/ turbine	Totals
High Pressure Gas Turbine	2	115	230
Low Pressure Gas Turbine	7	363	2,541
High Pressure Steam Turbine	2	243	486
High Pressure Steam Turbine	1	18	18
Low Pressure Steam Turbine	6	238	1,428
Low Pressure Steam Turbine	1	246	246
Totals	19		4,941

Economics - Comparison of Costs

Comparison of Proposed Process and <i>ABPIS</i> (Million\$US)		
	Proposed Process	<i>ABPIS</i>
Total ISBL with 40% Contingency	\$4,098.99	\$4,452.00
Total OSBL with 40% Contingency	\$7,220.60	\$2,086.00
Total Capital Cost	\$11,319.59	\$6,538.00
Total Instantaneous Costs (Location Adjustment, Licence Fees & Land Costs)	\$12,782.94	\$7,443.24
Total Investment Before Escalation (Owners Cost & Working Capital)	\$14,341.29	\$8,421.24
Cost of Escalation During Construction	\$1,462.00	\$1,462.00
TOTAL INVESTMENT (Million\$US)	\$15,803.29	\$9,983.07
Total Operating Costs (at 100% Operation)	\$7,009.95	\$3,228.80
Total Revenues (at 100% Operation)	\$6,740.47	\$5,645.67
TOTAL ANNUAL CASH FLOW AT 100% OPERATION	-\$269.47	\$2,416.87

Economics - Carbon Costs

Carbon Tax Evaluation (@ \$50/Mt)		
	Proposed Process	<i>ABPIS</i>
CO2 to Claus Plant	2,124	617
CO2 from Cat Cracker	1,037	2,760
CO2 Released by Rectisol	0	26,419
CO2 From Coal and Coke Burning	0	11,070
CO2 From Fuel Oil Burning	99	99
CO2 From Natural Gas Burning	56	127
TOTAL CO2 EMITTED (Mt/d)	3,316	41,092
TOTAL CARBON TAX (Million\$US/yr)	\$56	\$699

Economics: Plant as a Whole

Project Economics	
Parameter	DM2
Capital Cost per kW for the Power Plant (\$USD/KW)	\$1,470
OPEX per KWh for the Power Plant (\$USD/KWh)	\$0.11
Internal Rate of Return (%)	13.05%
Break-Even Carbon Tax (\$USD)	\$116

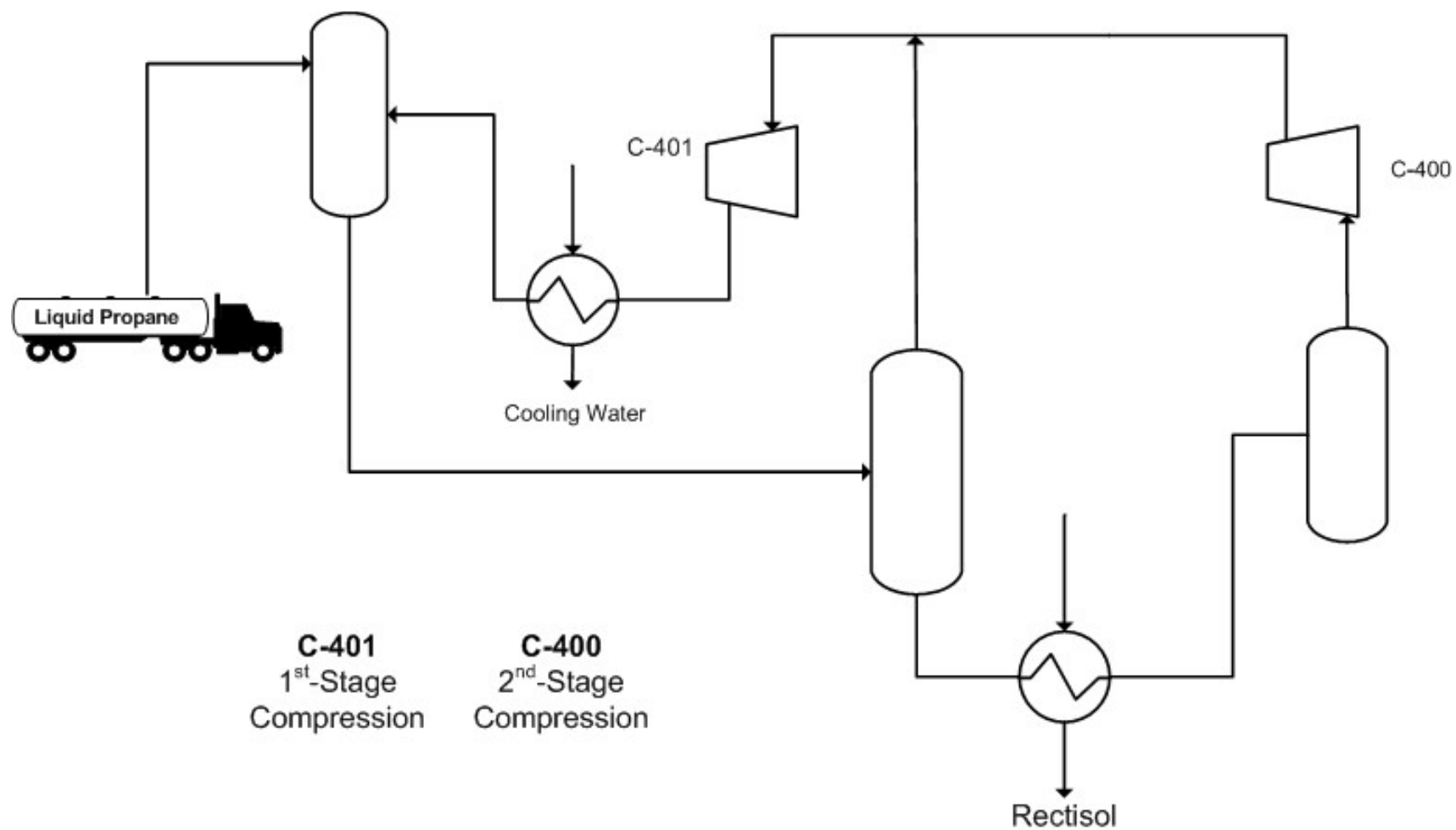
Profitability of Power Plant Alone

Project Economics: Power Plant Alone	
Parameter	DM2
Capital Cost per kW for the Power Plant (\$USD/KW)	\$1,470
Total Revenue	\$2,601
Total Operating Costs	\$3,727
Net Revenue	-\$1,126

Alberta Uses for Power

- This proposed process generates 4,185 MW of power
- ATCO Power, with other partners, has six new power plants under construction in Alberta that will provide enough energy to power 750,000 homes.
- ATCO Power also operates three coal-fired plants in the region with a capacity of over 1500 MW

Refrigeration Loop - PFD



Rectisol and Refrigeration

Rectisol Process		
Total Gas Flowrate In	320,900 lbmol/h	H₂ : 150,400 lbmol/h H₂S : 4,200 lbmol/h
Total Gas Flowrate Out	314,400 lbmol/h	H₂ : 150,400 lbmol/h H₂S : 0 lbmol/h
Total Methanol Flowrate	56,300 lbmol/h	

Refrigeration Loop	
Total Propane Flowrate	16,310 lbmol/h

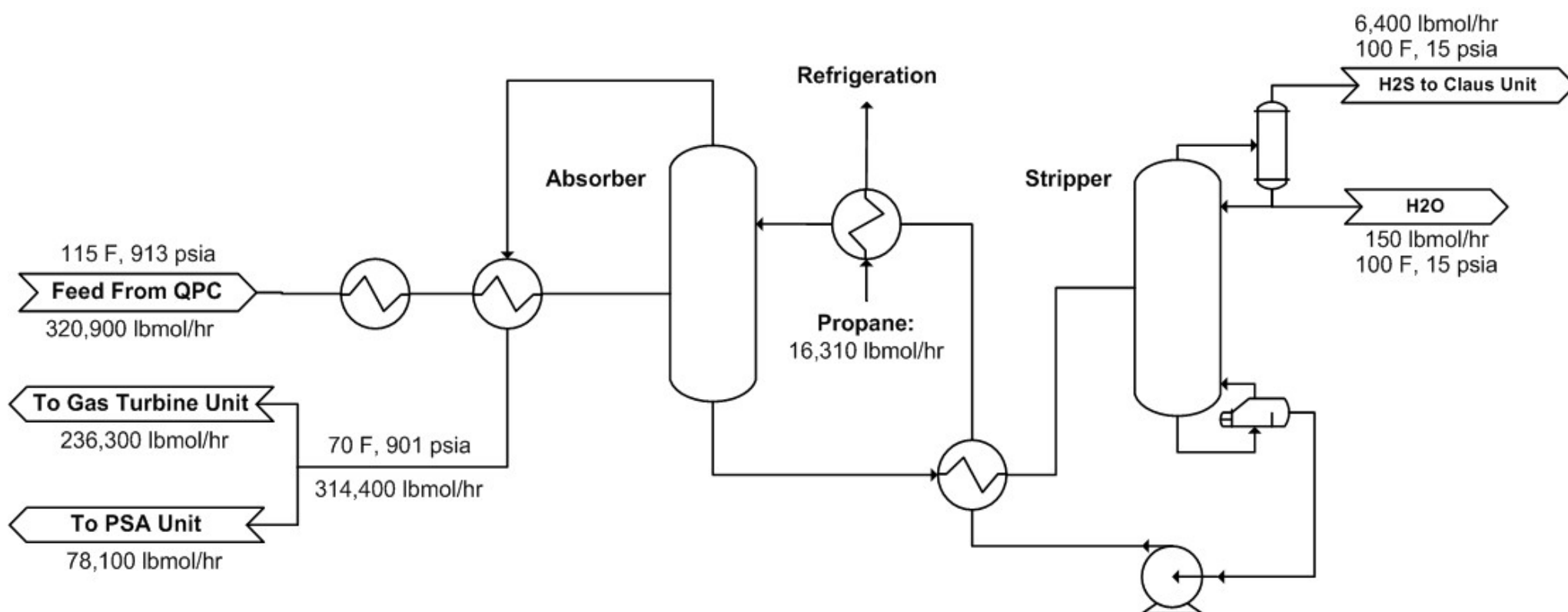
Refrigeration Loop

		C-400	C-401	E-302
- 40°F	Energy or Area	7,043 kW	10,540 kW	4,960.3 m ²
Cooling	Cost in 2006 \$'s	\$5,172,053	\$7,729,381	\$1,513,579
25°F	Energy or Area	4,020 kW	4,830 kW	24,653.4 m ²
Cooling	Cost in 2006 \$'s	\$2,961,359	\$3,553,705	\$7,459,928

Total Cost of -40F Cooling: \$14,415,000

Total Cost of 25F Cooling: \$13,970,000

Rectisol and Refrigeration - PFD



Absorber Top: 30 F, 903 psia
Absorber Bottom: 64 F, 908 psia

Stripper Top: 100 F, 15 psia
Stripper Bottom: 212 F, 17 psia